Sensor Data Cleaning and Correction: Application on the AirNow Fire and Smoke Map

Karoline Barkjohn[1], Amara Holder[1], Andrea Clements[1], Samuel Frederick[1,2], Ron Evans[3]

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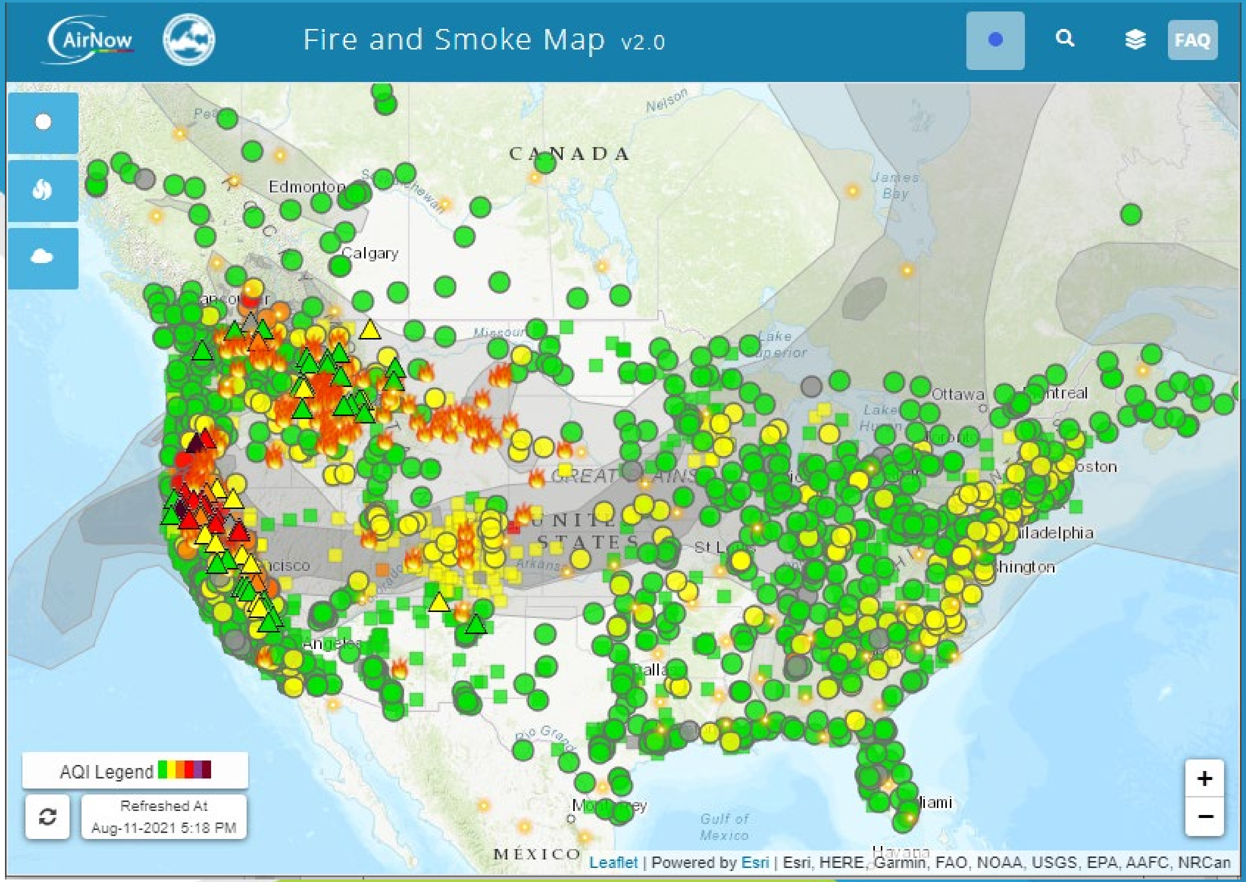
[2]Oak Ridge Associated Universities

[3]US EPA Office of Air Quality Planning and Standards



Fire.AirNow.gov

American Association for Aerosol Research

October 18-22, 2021

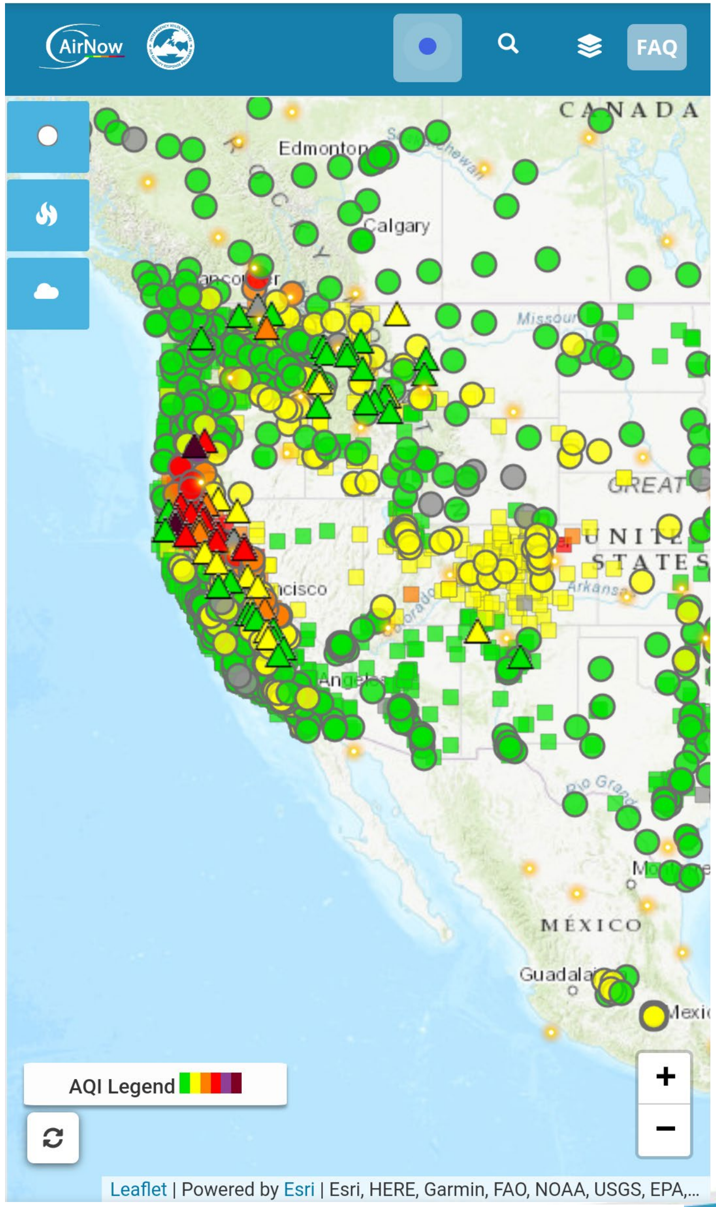
Office of Research and Development
Center for Environmental Measurement and Monitoring, Air Methods and Characterization Division

AirNow Fire & Smoke Map

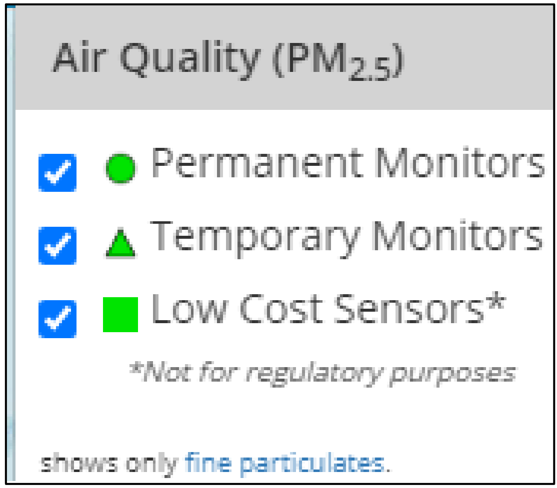
Objective: Provide enhanced air quality information critical during periods of wildland fires and smoke

- Merge multiple sources of information
- Higher time resolution data from sensors

Partnership between US Environmental Protection Agency and US Forest Service



Mobile AirNow Fire and Smoke map

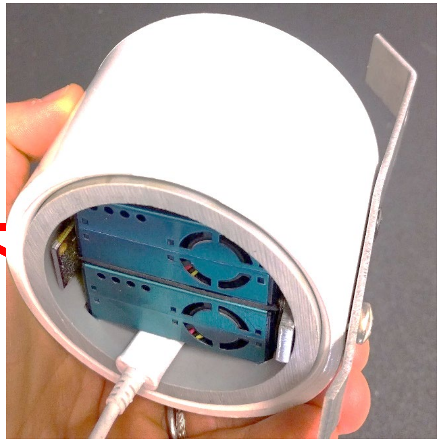


Sensors: Challenges and Rewards

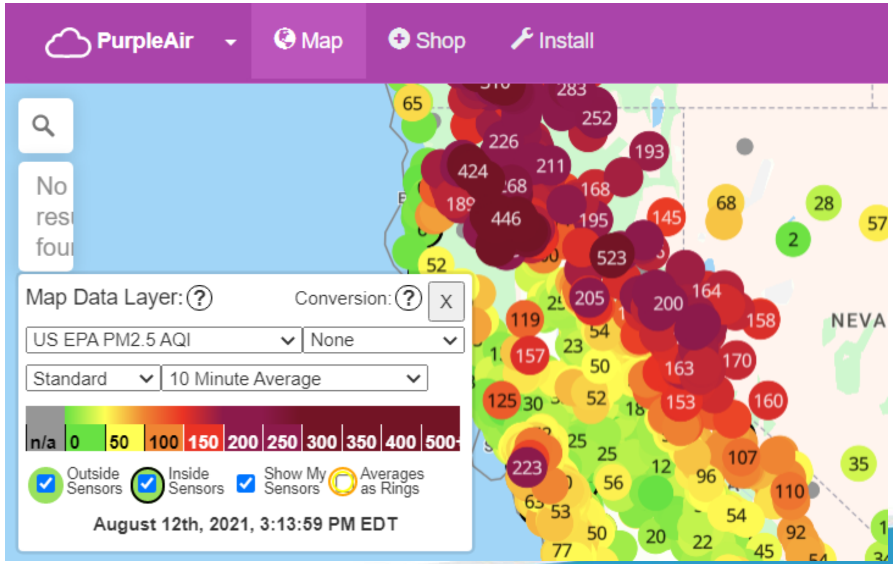
Challenges

- Data quality assurance methods needed
  - Data completeness
  - Outdoor sensors only
  - Exclusion when duplicate channels disagree
  - Correction required for bias and RH influence
- Communication: PurpleAir displays their information differently than the Fire and Smoke map

A & B channels



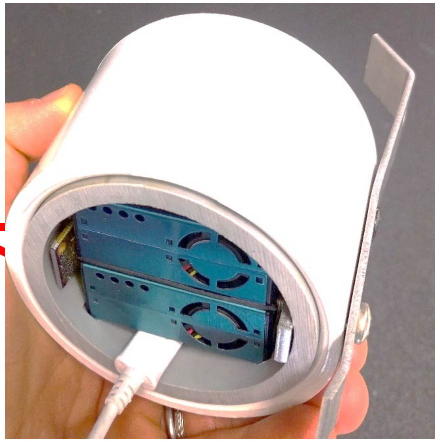
PurpleAir Sensor underside view



PurpleAir.com/map

Sensors: Challenges and Rewards

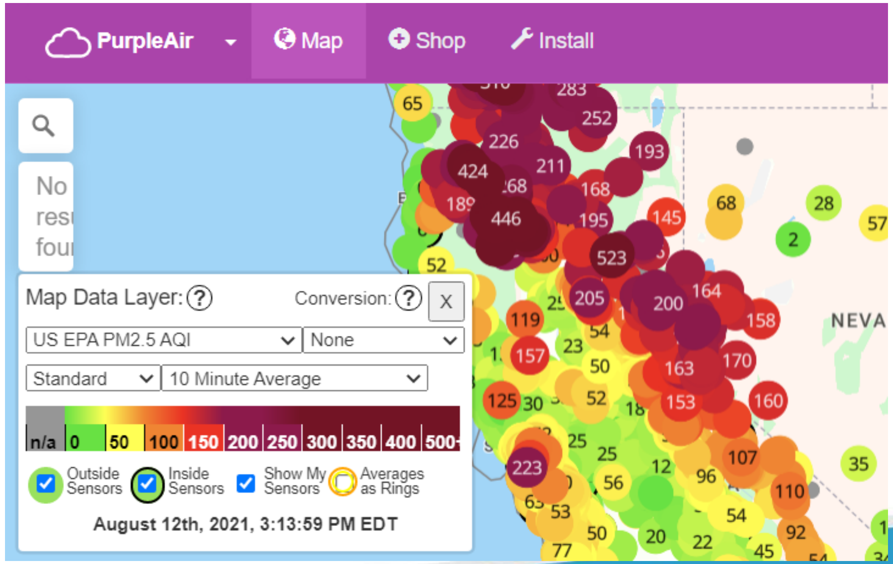
A & B channels



PurpleAir Sensor underside view

Rewards

- Add valuable cost-effective spatial information to the map
- Users can make decisions based on higher quality data from multiple sources



PurpleAir.com/map

2019
US-wide correction built

US-wide Correction Timeline

- Built and tested on 24-hr averaged data from Federal Reference and Equivalent methods (FRMs and FEMs)
- 16 States

Development and application of a United States-wide correction for $PM_{2.5}$ data collected with the PurpleAir sensor

Karoline K. Barkjohn[1], Brett Gantt[2], and Andrea L. Clements[1]

https://doi.org/10.5194/amt-14-4617-2021

Original US-wide correction

$$PM_{2.5} = 0.52*PA_{cf\_1} - 0.086*RH + 5.75$$

US-wide Correction Timeline

2019
US-wide correction built

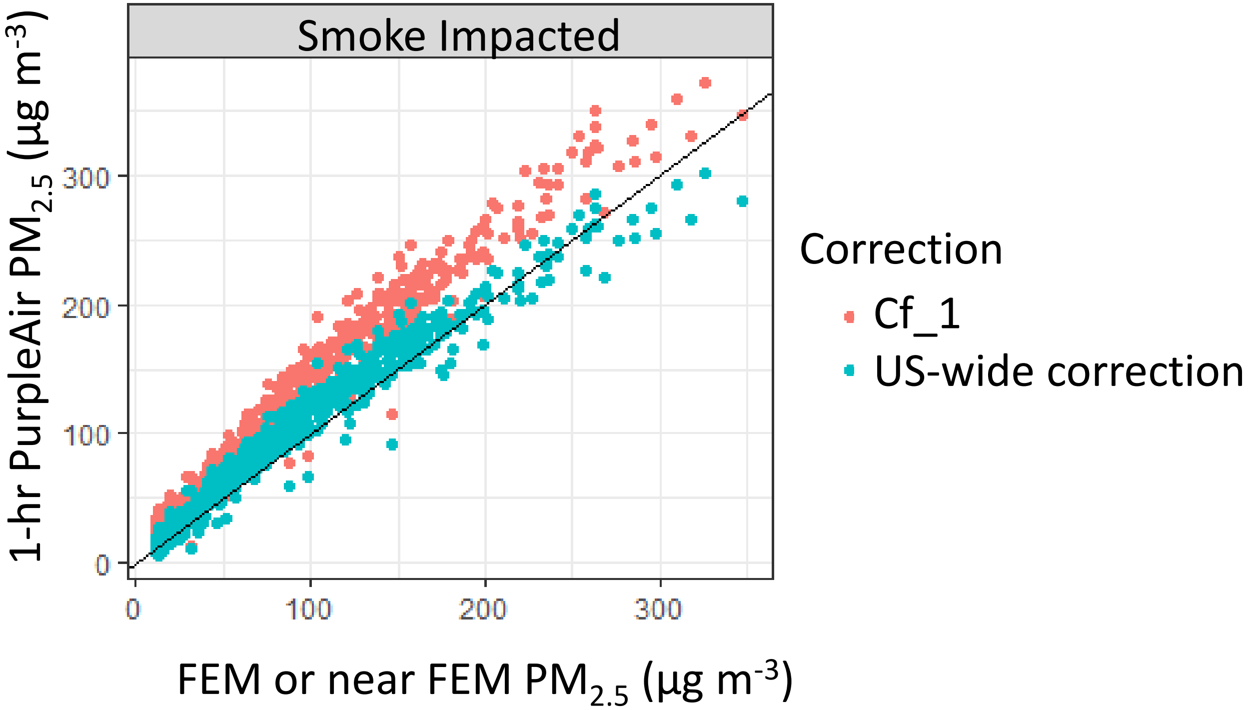
2020
Evaluated on smoke impacted datasets from 2018 & 2019

Tested on 1-hr smoke and ambient datasets

Article
Field Evaluation of Low-Cost Particulate Matter Sensors for Measuring Wildfire Smoke

Amara L. Holder [1,*], Anna K. Mebust [2], Lauren A. Maghran [2], Michael R. McGown [3], Kathleen E. Stewart [2], Dena M. Vallano [2], Robert A. Elleman [3] and Kirk R. Baker [4]

https://doi.org/10.3390/s20174796



US-wide Correction Timeline

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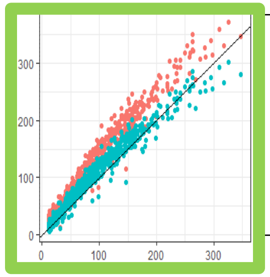
US-wide correction built

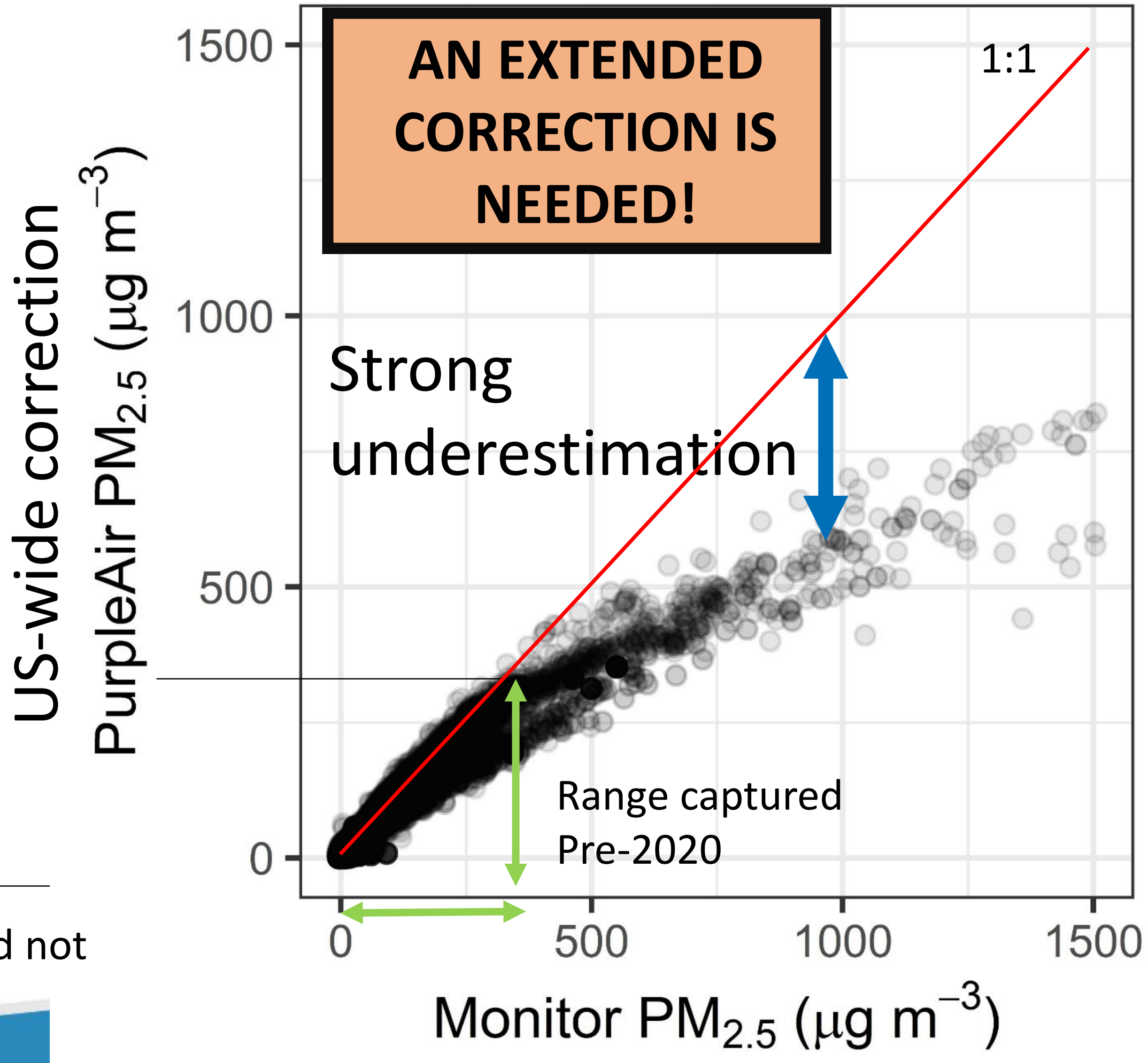
2020
Evaluated on smoke impacted datasets from 2018 & 2019

Summer 2020
Underpredicts at <u>extreme</u> smoke concentrations

Collocation data captured in 2020 spanned a much larger range of concentrations



Previous dataset did not capture full range



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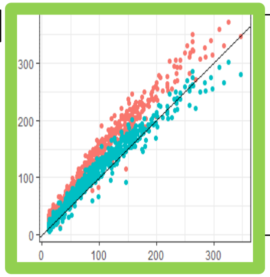
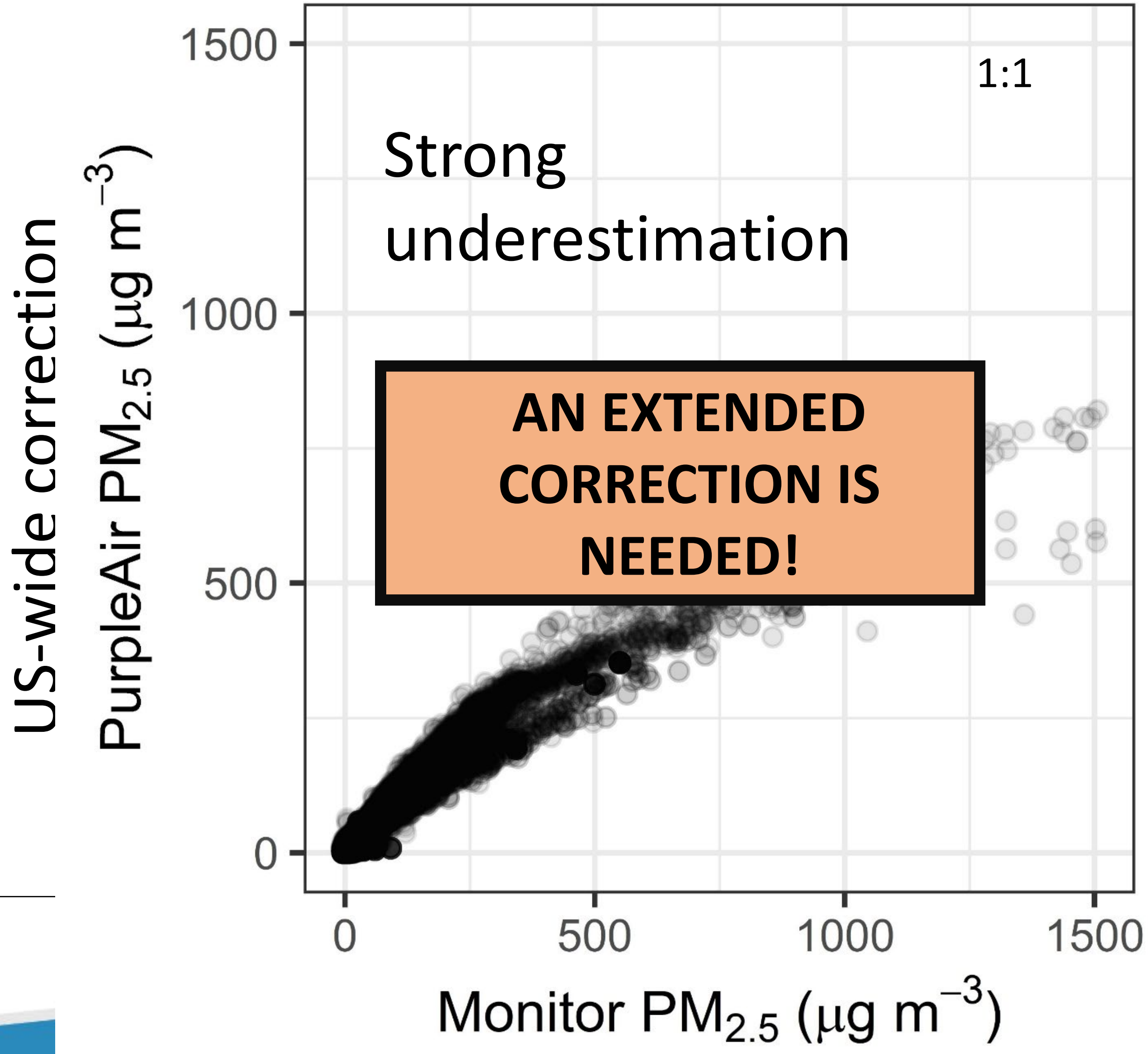




Dataset for Extended Correction

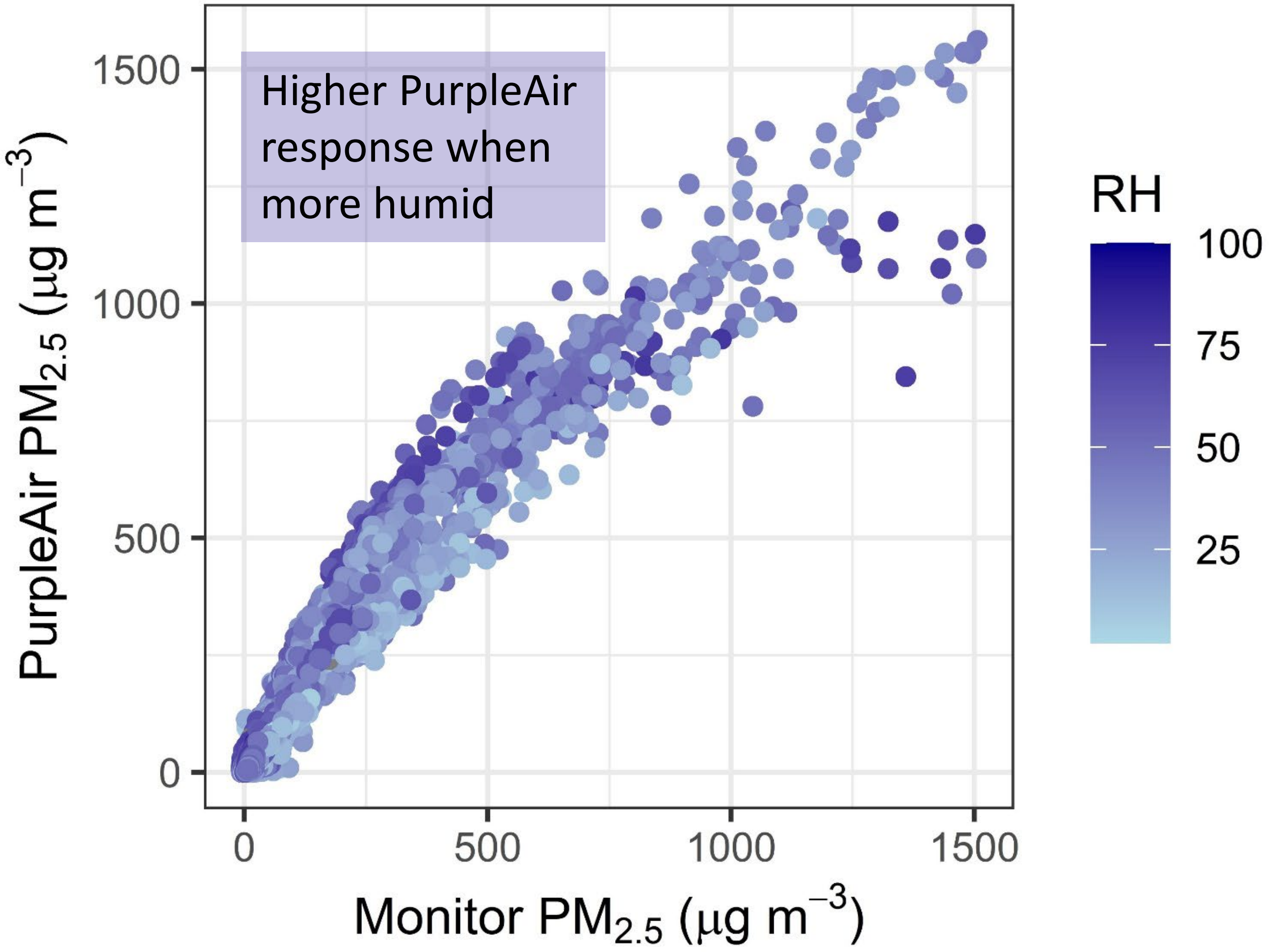
- Nearby sensor and monitor pairs identified during 2020 extreme smoke events (~>300 µg/m^3)
- Also included smoke and typical ambient collocations

- **Result:** Wide range of smoke and fire types and environmental conditions

Ambient
2018-2019 Smoke Impacted
2020 Extreme Smoke Impacted

Correction Requirements

- **Fits full range**
  - Important so that the map can be used during times of the year with and without smoke impacts
- **Considers relative humidity (RH) influence**
  - Important since monitors measure dry $PM_{2.5}$ and RH can increase light scattering per mass
- **Simple is better**
  - Want model to be broadly applicable and easy to interpret



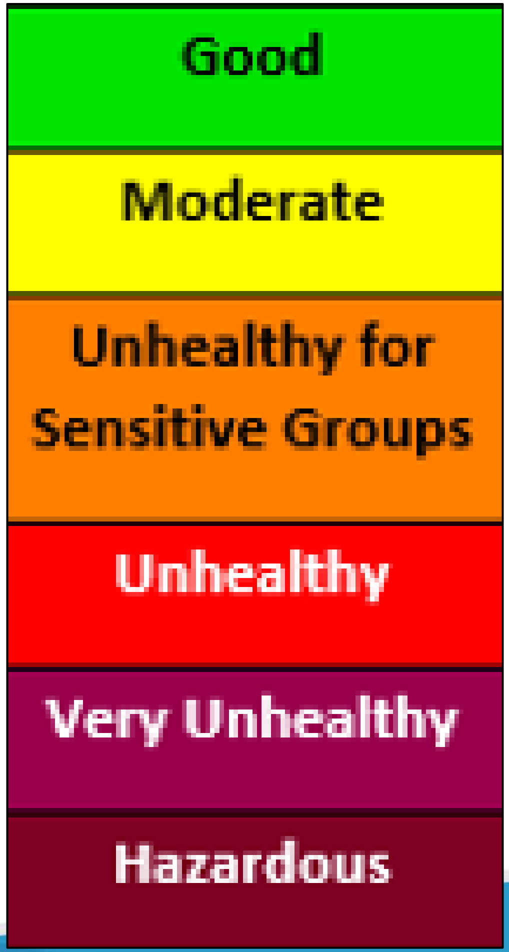
$PM_{2.5}$=Fine particulate Matter

Methods: Model Evaluation

- Evaluate performance at each AQI breakpoint
  - **AQI is the primary way risk is communicated on the map**
- Build and test using withholding

- Targets:
  - Bias* ≤ ± 5% in each bin
  - Reduce error† in each bin

Air Quality Index Categories



Bins to evaluate Performance

break points ±20%

10-14 μg m^{-3}

28-42 μg m^{-3}

44-66 μg m^{-3}

120-180 μg m^{-3}

200-300 μg m^{-3}

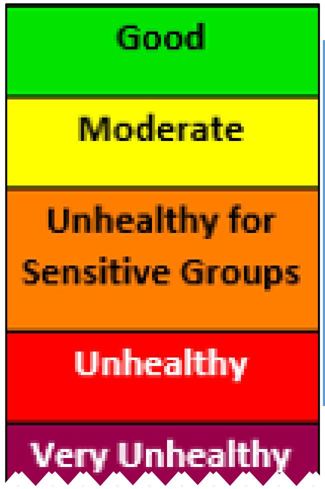
400-600 μg m^{-3} (calOSHA respirator)

*Normalized mean bias error (NMBE)
†Normalized mean absolute error (NMAE)

Final Correction

- Use the US-wide correction until PA_{cf_1} exceeds 343 μg m^{-3}

- **US-wide correction** best fit: 120-180 μg m^{-3}
  - Breakpoint: Unhealthy-Very Unhealthy
- **Quadradic fit** simplest best fit: 200-300 μg m^{-3}
  - Breakpoint: Very Unhealthy-Hazardous
- Equation change point selected so that it would fall between 2 bins and be less likely to influence the AQI category the data falls into



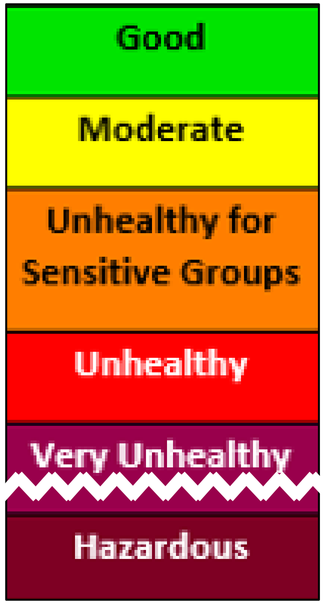
Low Concentration $PA_{cf\_1} \leq 343$ μg m^{-3} ~176-185 μg m^{-3} as measured by the corrected sensor	$PM_{2.5} = 0.52 \times PA_{cf\_1} - 0.086 \times RH + 5.75$

PA_{cf_1} = PurpleAir $PM_{2.5}$ from the higher of the 2 correction factors (cf) currently labeled as cf_1

Final Correction

- Use the US-wide correction until PA_{cf_1} exceeds 343 µg m^{-3}
- Then use a quadratic fit

- **US-wide correction** best fit: 120-180 µg m^{-3}
  - Breakpoint: Unhealthy-Very Unhealthy
- **Quadradic fit** simplest best fit: 200-300 µg m^{-3}
  - Breakpoint: Very Unhealthy-Hazardous
- Equation change point selected so that it would fall between 2 bins and be less likely to influence the AQI category the data falls into



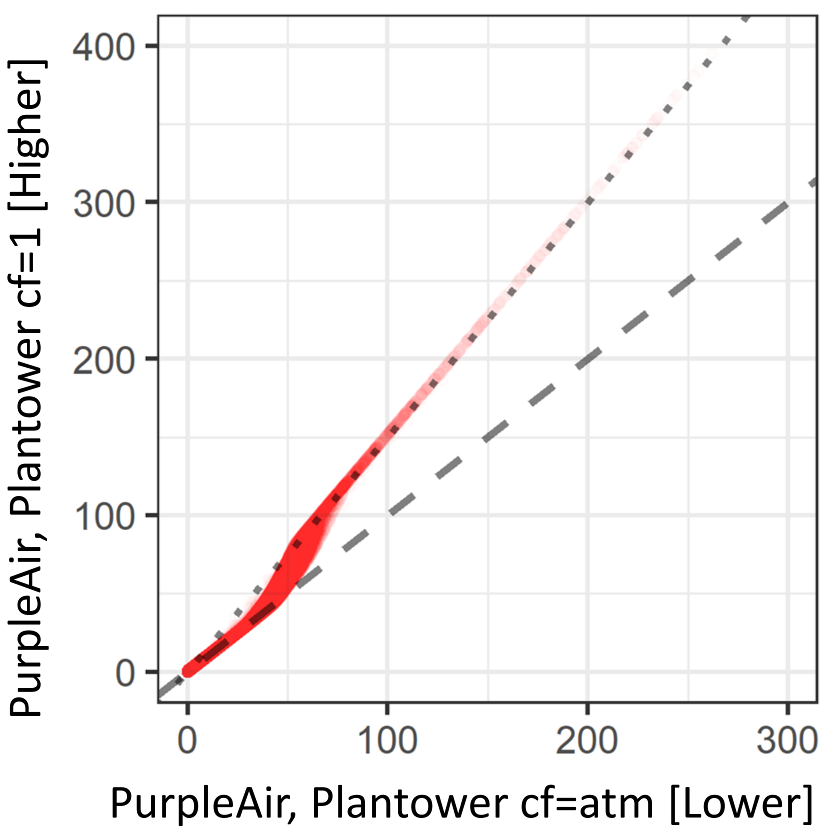
Low Concentration $PA_{cf\_1} \leq 343$ µg m^{-3} ~176-185 µg m^{-3} as measured by the corrected sensor	$PM_{2.5} = 0.52 \times PA_{cf\_1} - 0.086 \times RH + 5.75$
High Concentration $PA_{cf\_1} > 343$ µg m^{-3} ~207 µg m^{-3} as measured by the corrected sensor	$PM_{2.5} = 0.46 \times PA_{cf\_1} + 3.93 \times 10^{-4} \times PA_{cf\_1}{}^{2} + 2.97$

PA_{cf_1} = PurpleAir $PM_{2.5}$ from the higher of the 2 correction factors (cf) currently labeled as cf_1
*Correction also available for PA_{cf_atm} (the other cf) – currently in use due to API limitations

Current Fire and Smoke Map Correction (cf_atm)

- The PurpleAir US-wide & extended corrections were developed using **cf=1 [higher]**
  - Cf=1 is more strongly correlated with FRM/FEM/near FEM over the full concentration range
- If cf_atm is used due to real-time API limitations this piecewise equation may be used
  - In use on AirNow Fire and Smoke map (Summer 2021)

2-min comparison of cf_1 and cf_atm data



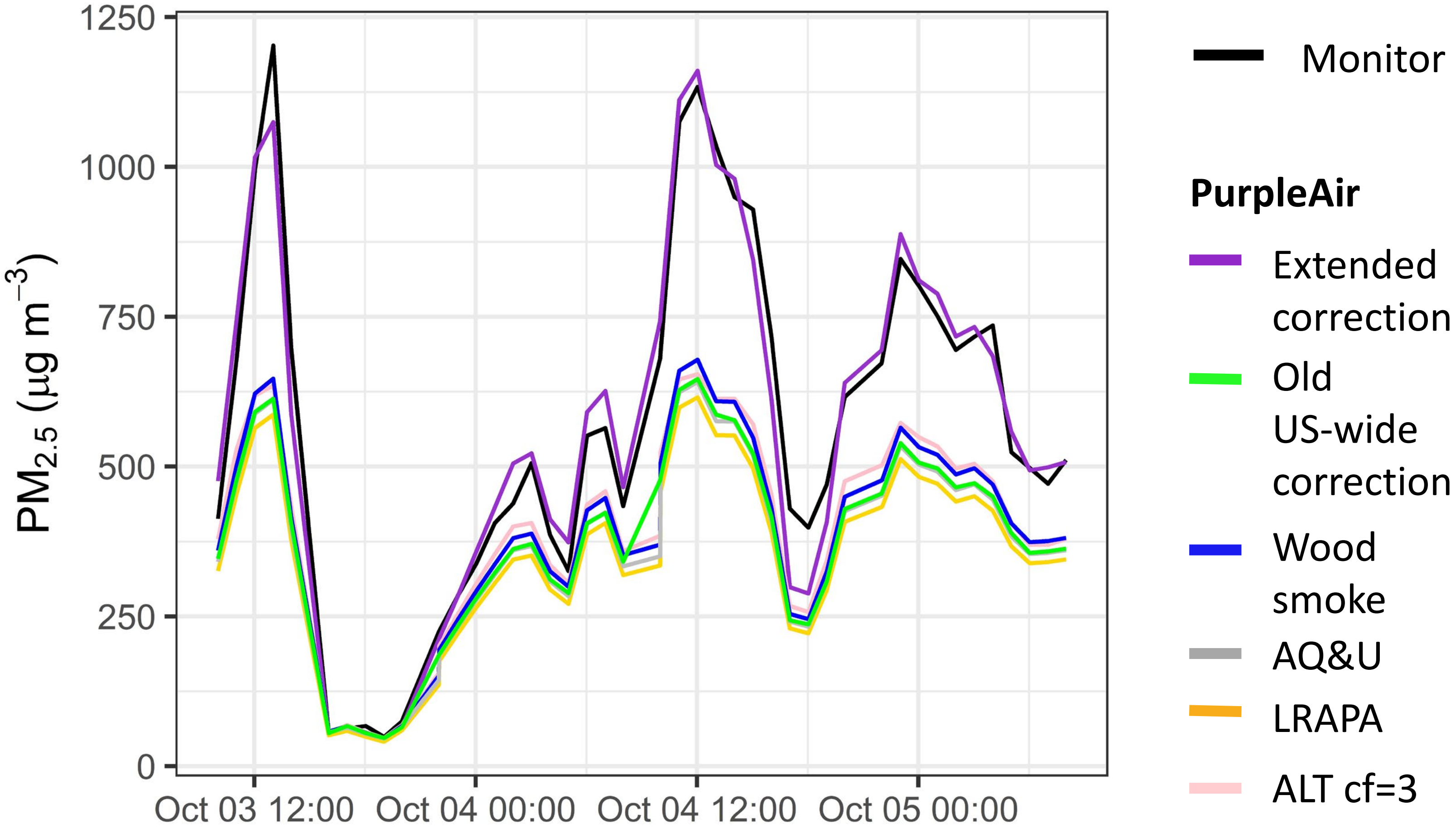
Adding in a transition equation that shifts the weight from one equation to the next

Low Concentration $PA_{cf\_atm} < 50\ \mu g\ m^{-3}$	$PM_{2.5} = 0.52 \times PA_{cf\_atm} - 0.086 \times RH + 5.75$
Mid Concentration $50\ \mu g\ m^{-3} \leq PA_{cf\_atm} < 229$	$PM_{2.5} = 0.786 \times PA_{cf\_atm} - 0.086 \times RH + 5.75$
High Concentration $PA_{cf\_atm} > 229\ \mu g\ m^{-3}$	$PM_{2.5} = 0.69 \times PA_{cf\_atm} + 8.84 \times 10^{-4} \times PA_{cf\_atm}^{2} + 2.97$

How does this compare to other corrections on the PurpleAir Map?

- Better agreement over the full range
- Other corrections look very similar to the old US-wide correction
- Confusing for users to know which correction to select on PurpleAir.com

Example: Forks of Salmon



[1]LRAPA – Lane Regional Air Protection Agency

Old US-wide correction almost entirely covers AQ&U

Team Effort

EPA Office of Air Quality Planning and Standards

- Ron Evans
- John White
- Phil Dickerson
- Lourdes Morales
- Michelle Wayland
- Rob Wildermann
- Alison Davis
- Susan Stone
- Kristen Benedict

EPA Office of Research and Developement

- Amara Holder
- Andrea L. Clements
- Samuel Frederick (Student Services Contractor)

US Forest Service AirFire

- Sim Larkin
- Stuart Illson (University of Washington)
- Jonathan Callahan (Mazama Science)

US Forest Service

- Pete Lahm

This work would not have been possible without support from partner state, tribal and local agencies, EPA regional offices and other federal agencies including the National Park Service, and the Wildland Fire Air Quality Response Program.

Resources & Publications

Additional details about EPA's work with air sensors for the Fire & Smoke map

- https://www.epa.gov/air-sensor-toolbox/technical-approaches-sensor-data-airnow-fire-and-smoke-map

EPA Tools & Resources Public Webinar (May 19th 2021):

https://www.epa.gov/research-states/epa-tools-and-resources-webinar-series

- Includes section on sensor users frequently asked questions

Contact:
Barkjohn.Karoline@epa.gov

Project Publications:

Holder, A., A. Mebust, L. Maghran, M. McGown, K. Steward, D. Vallano, R. Elleman, and K. Baker, 2020. 'Field Evaluation of Low-Cost Particulate Matter Sensors for Measuring Wildfire Smoke', Sensors. https://doi.org/10.3390/s20174796

Barkjohn, K, B. Gantt, A. Clements, 2020 'Development of a United States Wide Correction for $PM_{2.5}$ Data Collected with the PurpleAir Sensor', Atmospheric Measurement Techniques Discussion. https://doi.org/10.5194/amt-2020-413

Barkjohn, K, A. Holder, S. Frederick, A. Clements, (in preparation) 'PurpleAir $PM_{2.5}$ US Correction and Performance During Smoke Events'.

Additional slides

Reducing gap in cf_atm PurpleAir correction equations

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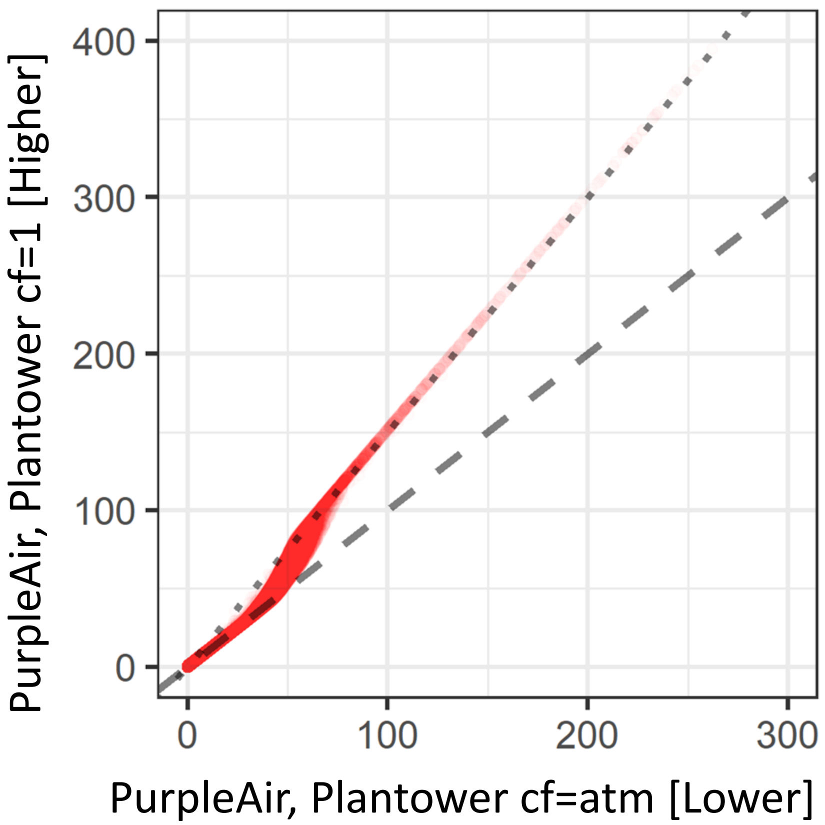
Current Correction (cf_atm)

- The PurpleAir US-wide & extended corrections were developed using **cf=1 [higher]**
  - Cf=1 is more strongly correlated with FRM/FEM/near FEM over the full concentration range
- PurpleAir only provides averaged (e.g. 10-min to daily) cf_atm data in their new API
  - Cf_atm equation in use on AirNow Fire and Smoke map (September 2021)

2-min comparison of cf_1 and cf_atm data



Add in a transition equation that shifts the weight from one equation to the next

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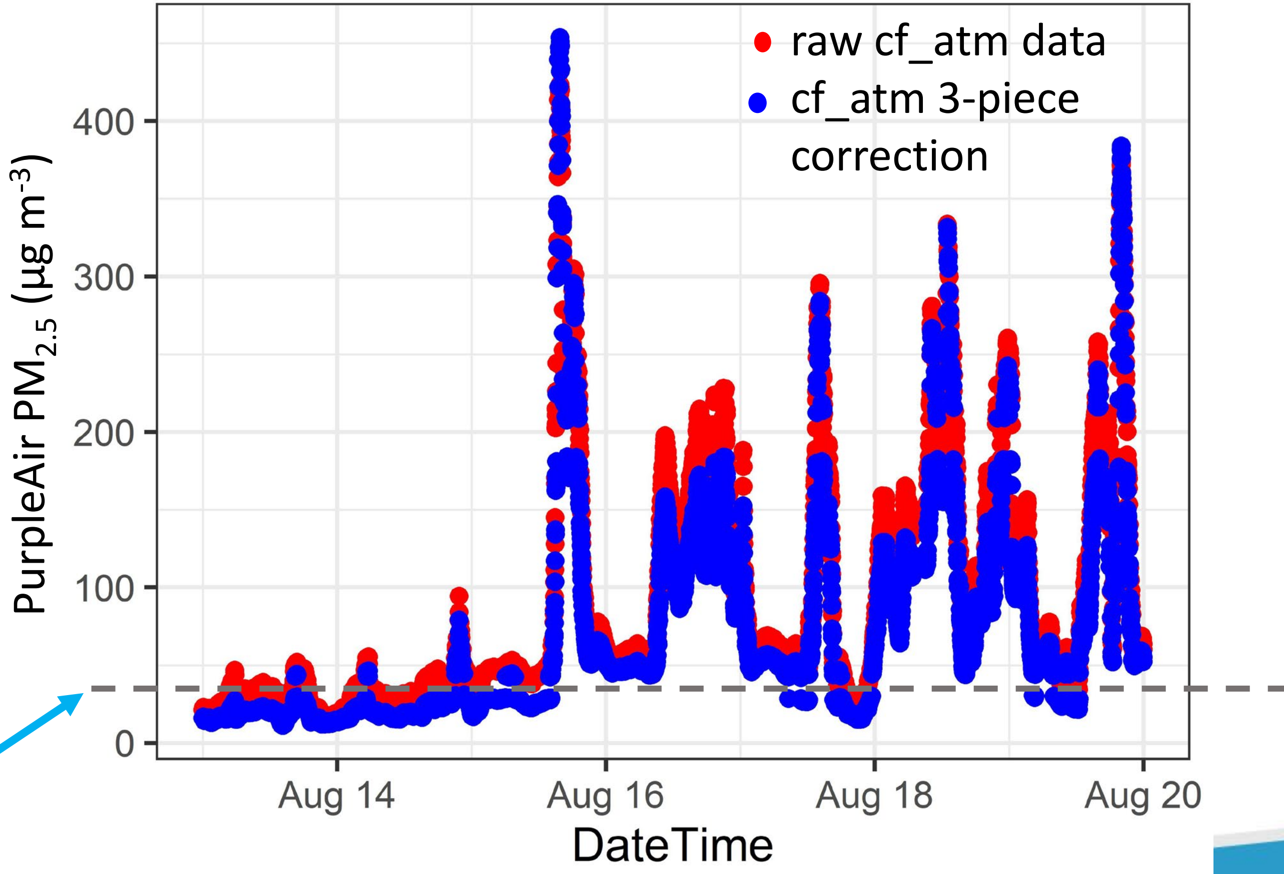
Challenge

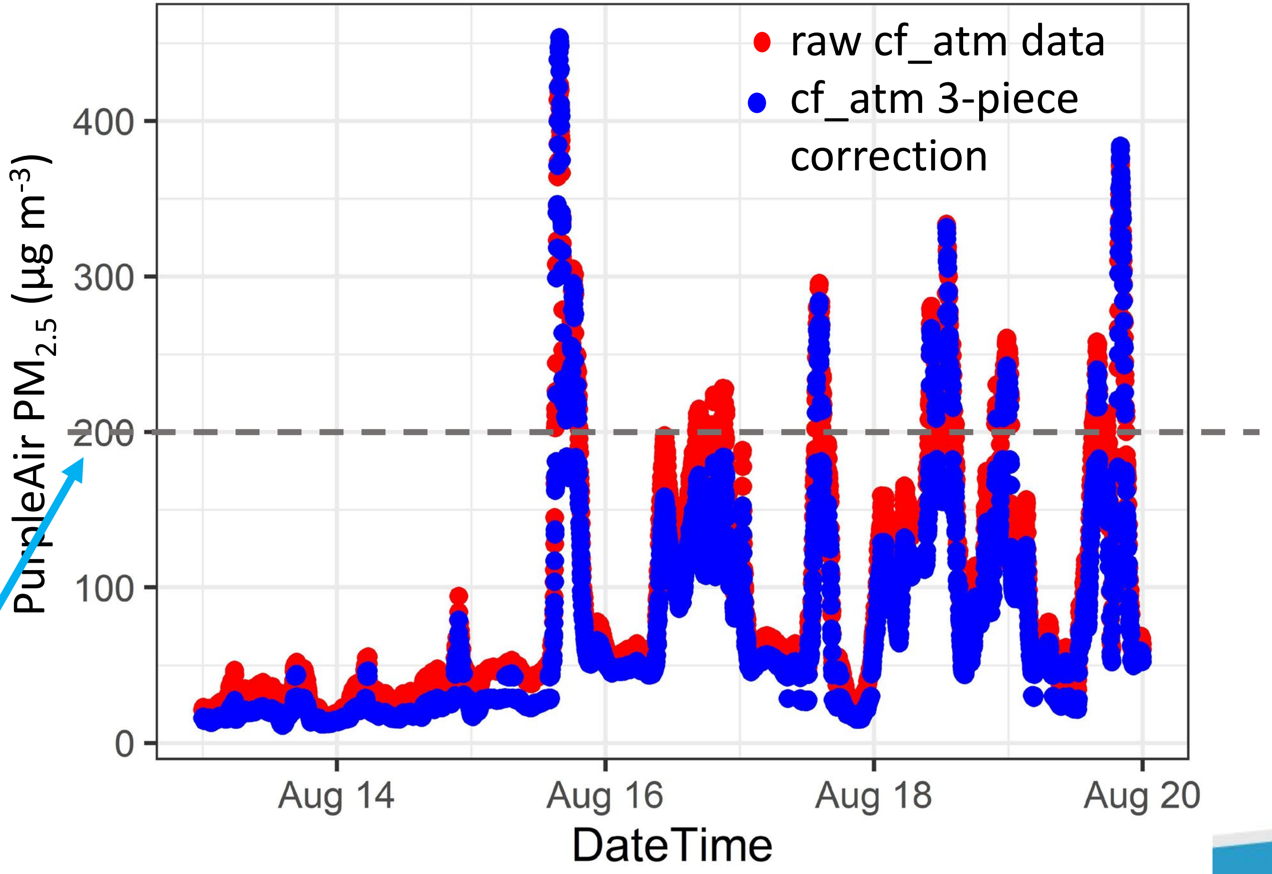
- A piecewise equation can cause discontinuity leading to large jumps in concentration when a measurement exceeds the threshold



In this gap a <1 µg m^{-3} change in cf_atm concentration leads to a jump of ~ 13 µg m^{-3}

Challenge

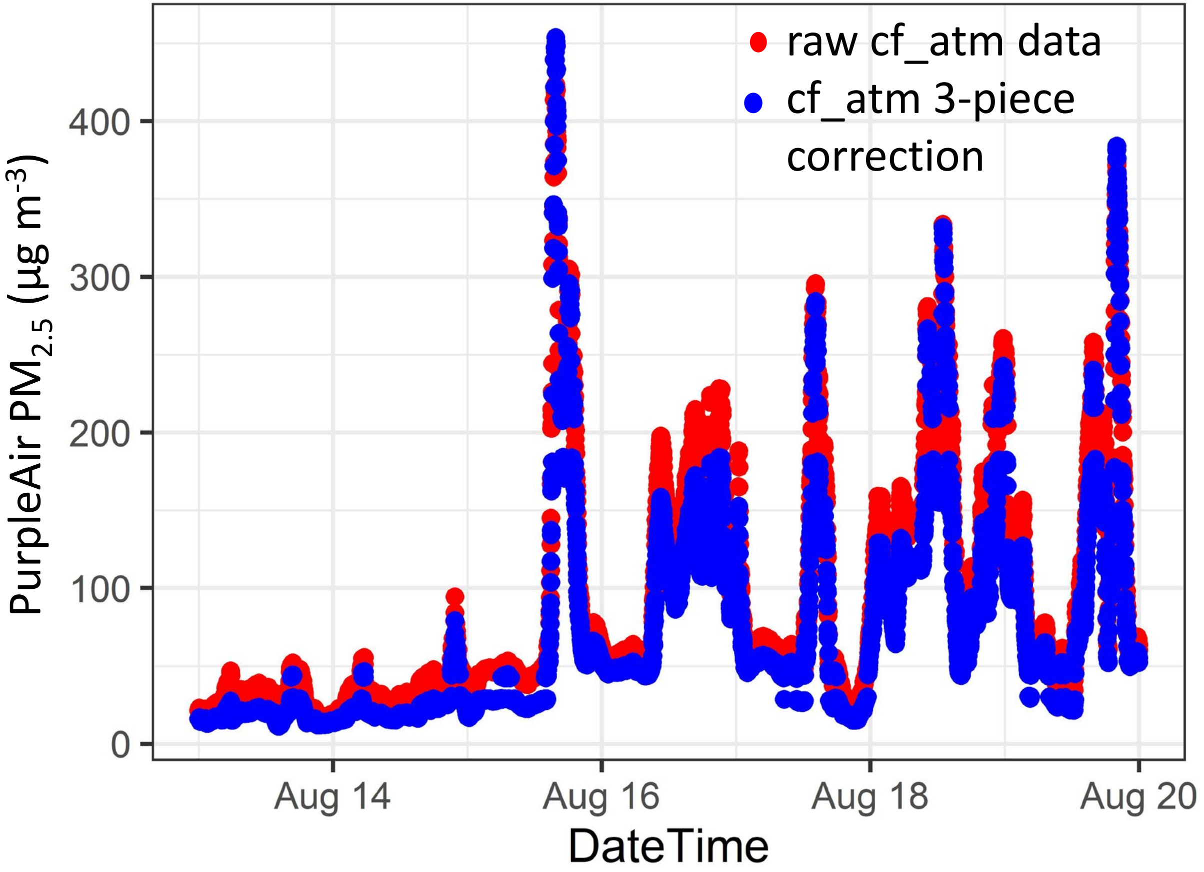
- A piecewise equation can cause discontinuity leading to large jumps in concentration when a measurement exceeds the threshold



The second gap is less visible since the concentrations are much higher but could be improved

Challenge

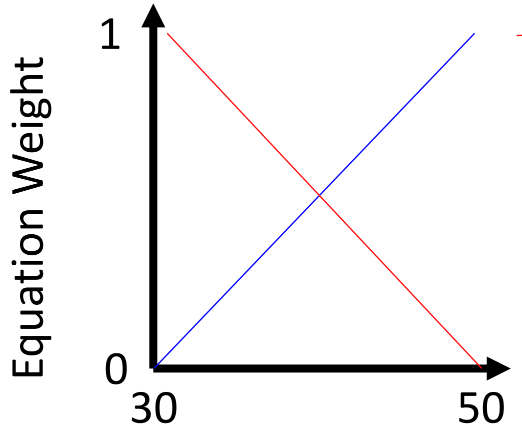
- A piecewise equation can cause discontinuity leading to large jumps in concentration when a measurement exceeds the threshold
- Initially a sigmoid was proposed but was too computationally expensive
- Solution: Use a **5-piece equation** with transitions between the 3 pieces



Transition from Low to Mid

1. Low Concentration PA_{cf_atm} < 50 µg m^{-3}	$PM_{2.5}$ = 0.52 x PA_{cf_atm} - 0.086 x RH + 5.75
2. Mid Concentration 50 µg m^{-3}≤PA_{cf_atm} <229	$PM_{2.5}$ = 0.786 x PA_{cf_atm} - 0.086 x RH + 5.75

- **30 ≤ x <50** µg/m^3
- Y = (0.786*(x/20 - 3/2)
  + 0.524*(1 - (x/20 - 3/2)))*x
  - 0.0862*RH + 5.75



Weight Equation 1
$$Y = \frac{1}{20}x - \frac{30}{20}$$

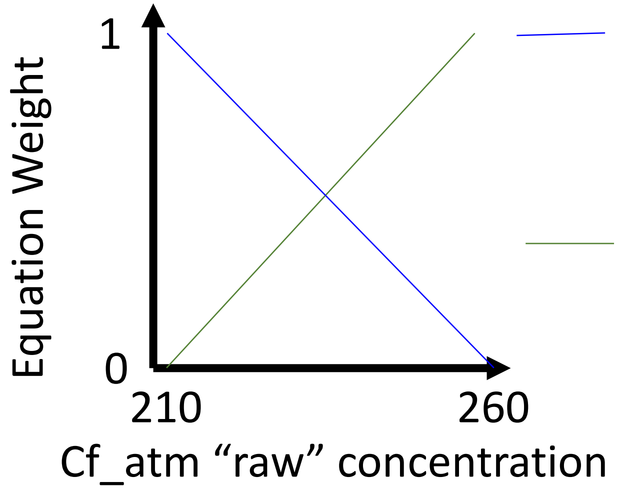
Weight Equation 2
$$Y = 1 - (\frac{1}{20}x - \frac{30}{20})$$

- Raw: 30, 50 µg/m^3
- Corrected: low: 13-21 µg/m^3 , high: 36-45 µg/m^3 (range since RH dependent)

Transition from Mid to High

2. Mid Concentration 50 μg $m^{-3} \le PA_{cf\_atm} < 229$	$PM_{2.5} = 0.786 \times PA_{cf\_atm} - 0.086 \times RH + 5.75$
3. High Concentration $PA_{cf\_atm} > 229$ μg m^{-3}	$PM_{2.5} = 0.69 \times PA_{cf\_atm} + 8.84 \times 10^{-4} \times PA_{cf\_atm}^{2} + 2.97$

- **210 ≤ x <260** μg/m^3
- Y = (0.69*(x/50 – 21/5)
  + 0.786*(1 - (x/50 – 21/5)))*x
  -0.0862*RH*(1 - (x /50 – 21/5))
  + 2.966*(x/50 – 21/5)
  + 5.75*(1 - (x/50 – 21/5))
  + 8.84*(10^{-4})*x^{2}*(x/50 – 21/5)



Weight Equation 2

$$Y = \frac{1}{50}x - \frac{210}{50}$$

Weight Equation 3

$$Y = 1 - \left(\frac{1}{50}x - \frac{210}{50}\right)$$

- Raw: 210, 260 μg/m^3
- Corrected: low: 162-171 μg/m^3 (RH dependent), high: 242 μg/m^3
- 150 and 250 μg/m^3 are the surrounding AQI breakpoints
  - Transition from ~10 μg/m^3 above the lower limit to 10 μg/m^3 below the higher limit

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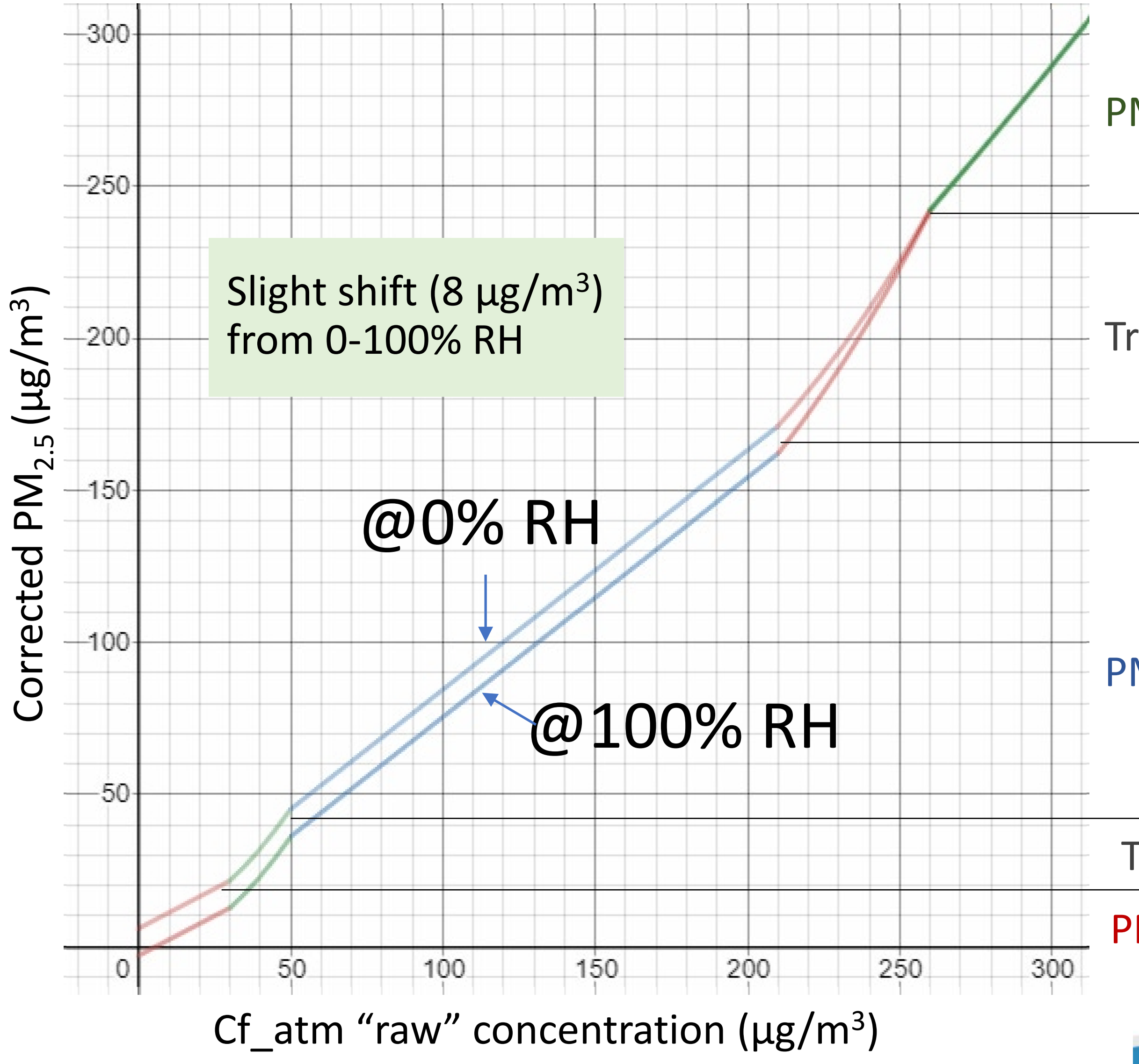
Full Equation

- y={**0 ≤ x <30**: 0.524*x - 0.0862*RH + 5.75}
- y={**30≤ x <50**: (0.786*(x/20 - 3/2) + 0.524*(1 - (x/20 - 3/2)))*x -0.0862*RH + 5.75}
- y={**50 ≤ x <210**: 0.786*x - 0.0862*RH + 5.75}
- y={**210 ≤ x <260**: (0.69*(x/50 – 21/5) + 0.786*(1 - (x/50 – 21/5)))*x - 0.0862*RH*(1 - (x/50 – 21/5)) + 2.966*(x/50 – 21/5) + 5.75*(1 - (x/50 – 21/5)) + 8.84*(10^{-4})*x^{2}*(x/50 – 21/5)}
- y={**260 ≤ x**: 2.966 + 0.69*x + 8.84*10^{-4}*x^2}

- y= corrected $PM_{2.5}$ $\mu g/m^3$
- x= $PM_{2.5}$ cf_atm (lower)
- RH= Relative humidity as measured by the PurpleAir

Full Equation by RH



$PM_{2.5} = 0.69 \times PA_{cf\_atm} + 8.84 \times 10^{-4} \times PA_{cf\_atm}{}^{2} + 2.97$

Transition

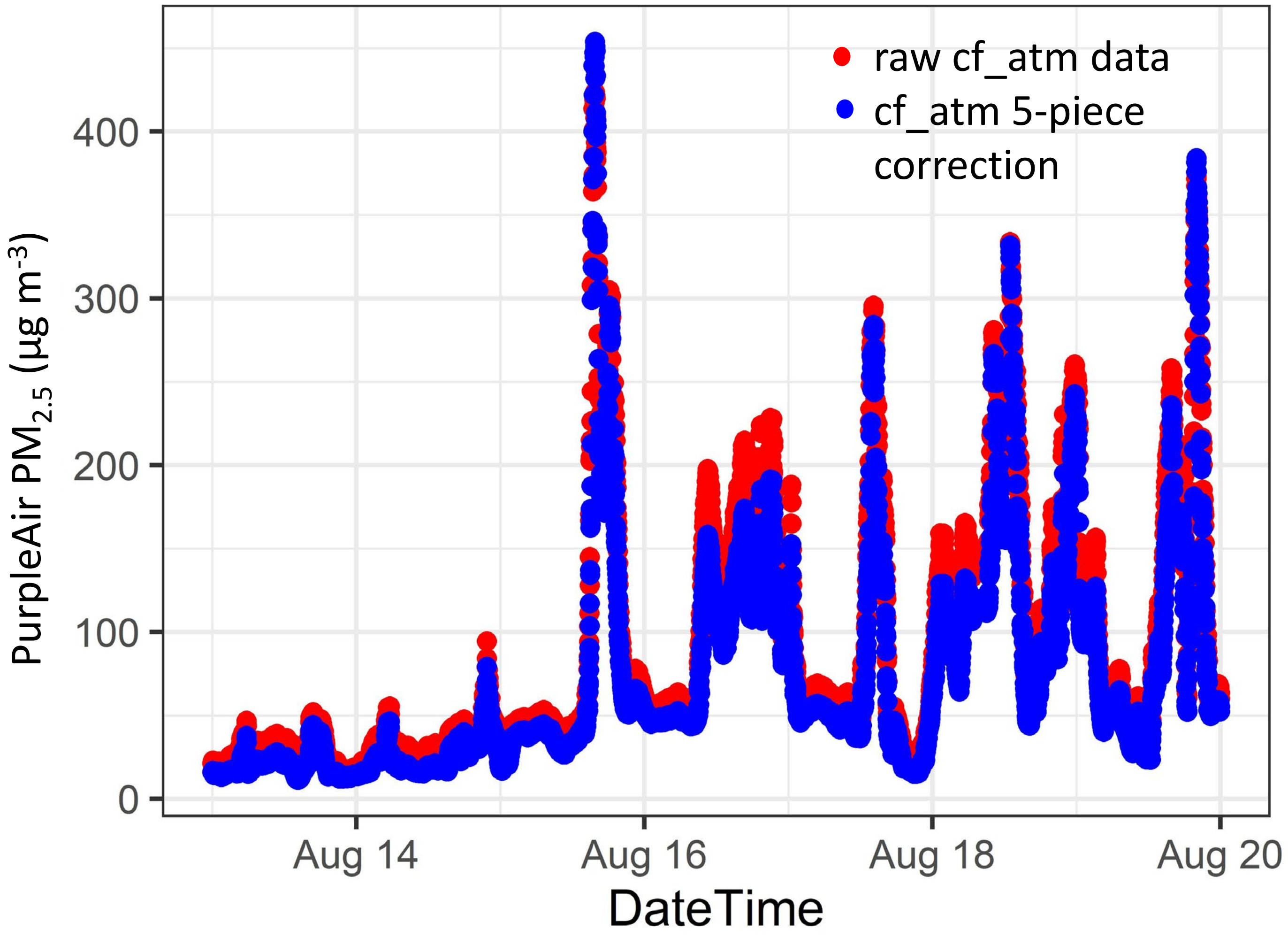
$PM_{2.5} = 0.786 \times PA_{cf\_atm} - 0.086 \times RH + 5.75$

Transition

$PM_{2.5} = 0.52 \times PA_{cf\_atm} - 0.086 \times RH + 5.75$

Initial Time Series with 5-piece Equation

- No longer the same distinct gaps/jumps



Evaluate 5-piece equation on Phoenix & Forks of Salmon Datasets

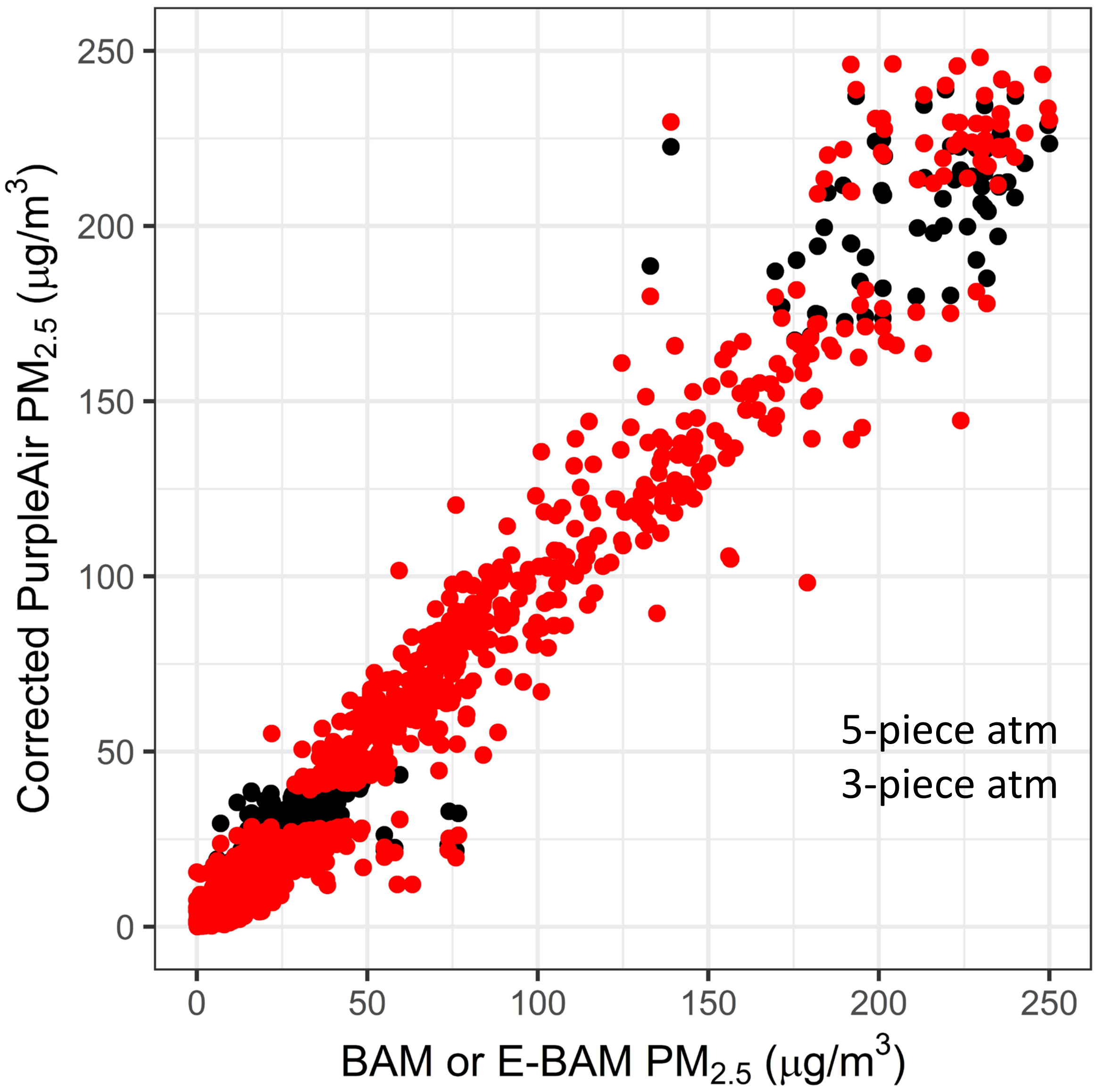
Hourly averaged collocations

~1 year in Phoenix with a BAM

~2 months extreme smoke in Forks of Salmon with an E-BAM

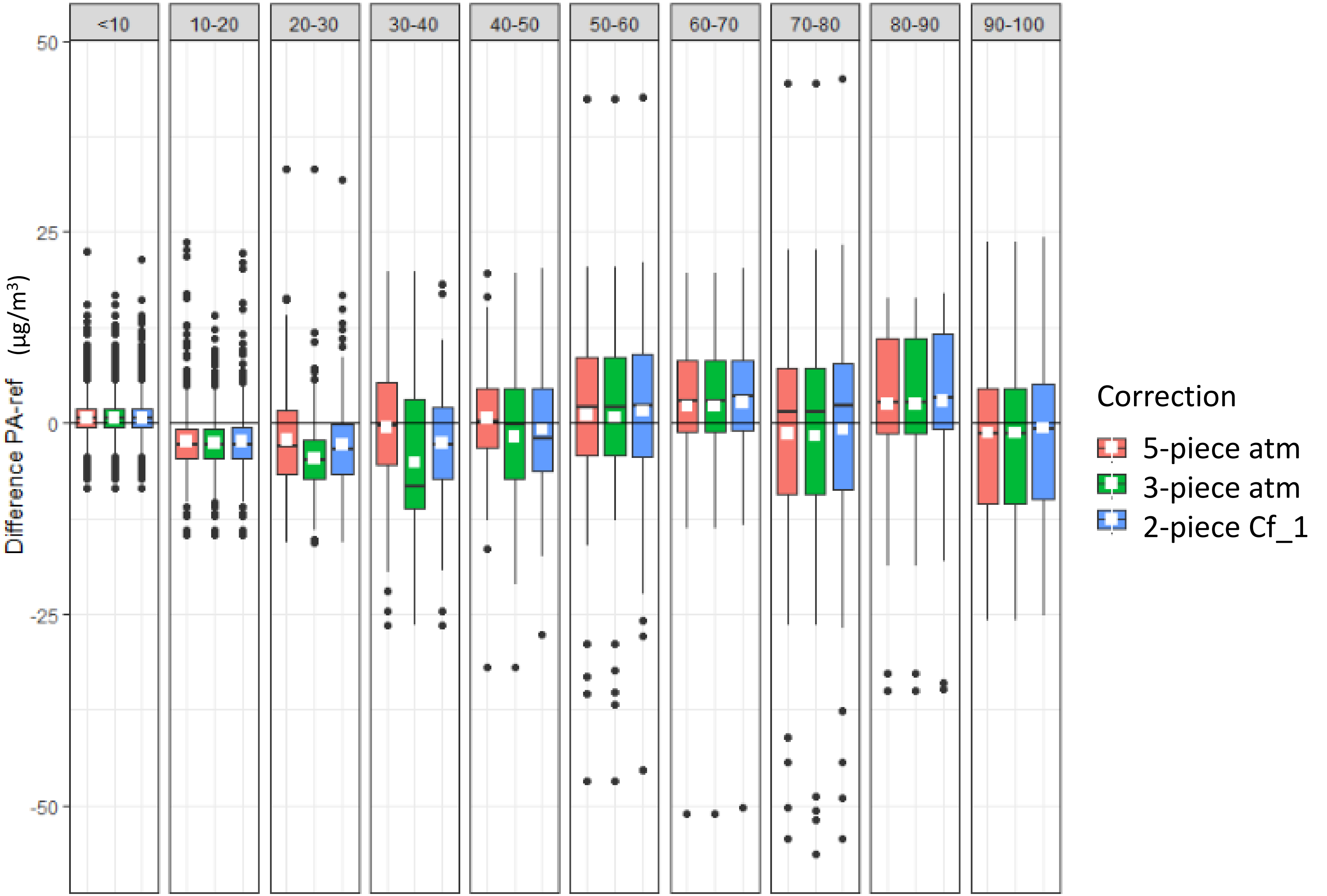
Performance Change

- Transition zones in the 5-piece equation
  - Remove low concentration gap
  - Remove high concentration gap



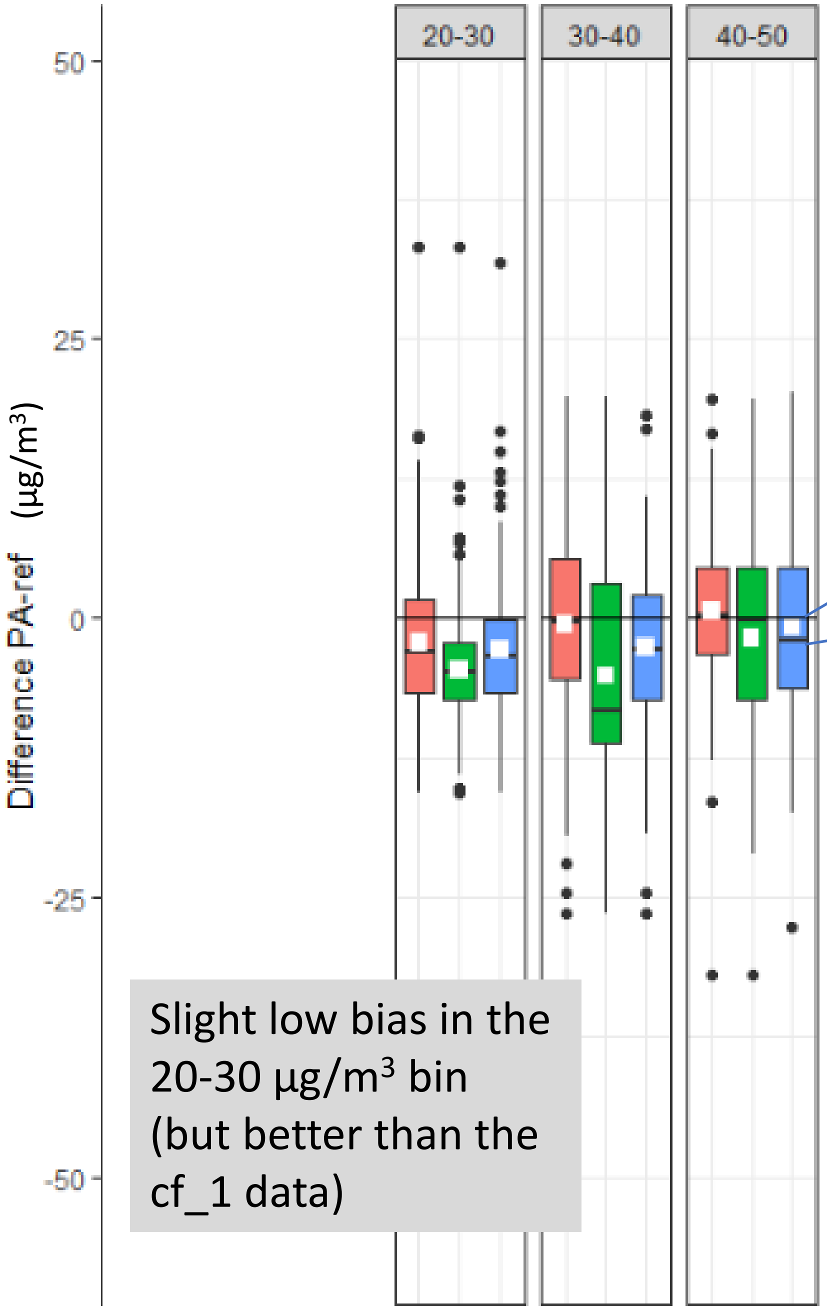
Performance By Concentration

- 5-piece performs more similarly to the cf_1
- Ref
  - E-BAM in Forks of Salmon
  - BAM in Phoenix



Performance By Concentration

- 5 piece performs more similarly to the cf_1
- Ref
  - E-BAM in Forks of Salmon
  - BAM in Phoenix



Bins of interest near the transition

Mean (white box)

Median (center line of boxplot)

Slight low bias in the 20-30 µg/m^3 bin (but better than the cf_1 data)

Conclusion

5-piece equation (i.e. 3-piece equation with 2 transition zones)

- Reduces bias in the 20-50 µg/m³ range
- Reduces gaps/jumps induced where the original 3 pieces meet
- **Will be implemented on the Fire and Smoke Map ~October 2021**